Theory of the Redundancy Inflation Error (RIE): bounds, edge cases and a scale-free decomposition

CRITCLOUD repair — supporting material

All propositions below are verified numerically by `check_rie_theory.py` (2,000 random weight vectors and correlation matrices, plus the QWS correlation matrix).

1 Definitions

Let $\mathbf{w} \in \Delta^{n-1} = \{\mathbf{w} : w_j \geq 0,\ \sum_j w_j = 1\}$ be a weight vector and $R = [r_{jk}]$ a correlation matrix of $n \geq 2$ criteria. Write $\tilde{R} = |R| - I$ (absolute correlations with a zero diagonal) and $\mathbf{g} = \tilde{R}\mathbf{w}$, i.e. $g_i = \sum_{l \neq i} w_l |r_{il}|$.

Definition 1 (RIE)**.** $\mathrm{RIE}(\mathbf{w}) = \sum_{j<k} w_j w_k |r_{jk}| = \frac{1}{2}\, \mathbf{w}^\top \tilde{R}\, \mathbf{w}$.

Definition 2 (Concentration measures)**.** $\mathrm{HHI}(\mathbf{w}) = \sum_j w_j^2$, effective number of criteria $N_{\mathrm{eff}} = 1/\mathrm{HHI}$, Shannon entropy $H(\mathbf{w}) = -\sum_j w_j \log w_j$, and $\mathrm{RIE}_{\mathrm{adj}} = \mathrm{RIE}/H$.

Definition 3 (Weighted Mean Redundancy)**.** For $\mathrm{HHI}(\mathbf{w}) < 1$,

$$\mathrm{WMR}(\mathbf{w}) = \frac{\sum_{j \neq k} w_j w_k |r_{jk}|}{\sum_{j \neq k} w_j w_k}.$$

WMR is the expected absolute correlation between two *distinct* criteria drawn independently with probabilities $\mathbf{w}$.

2 Results

Proposition 1 (Bounds)**.** *For every* $\mathbf{w} \in \Delta^{n-1}$*,*

$$0 \ \leq\ \mathrm{RIE}(\mathbf{w}) \ \leq\ \tfrac{1}{2}\big(1 - \mathrm{HHI}(\mathbf{w})\big) \ \leq\ \frac{n-1}{2n}.$$

The first inequality is an equality iff $w_j w_k |r_{jk}| = 0$ *for all* $j \neq k$*; the second iff* $|r_{jk}| = 1$ *for every pair with* $w_j w_k > 0$*; the third iff* $\mathbf{w} = \mathbf{1}/n$*.*

Proof. All terms $w_j w_k |r_{jk}|$ are non-negative, which gives the lower bound. Since $|r_{jk}| \leq 1$, $2\,\mathrm{RIE} \leq \sum_{j \neq k} w_j w_k = (\sum_j w_j)^2 - \sum_j w_j^2 = 1 - \mathrm{HHI}$. Finally $\mathrm{HHI} \geq 1/n$ by the Cauchy–Schwarz inequality, with equality only at the uniform vector. □

Remark 1. The original manuscript's "Property 2" ($\mathrm{RIE}_{\max} = (n-1)/(2n)$ at equal weights and $|r_{jk}| = 1$) is the special case of the last two inequalities. The bound $\frac{1}{2}(1 - \mathrm{HHI})$ is sharper because it holds for every $\mathbf{w}$.

Proposition 2 (Exact effect of moving weight)**.** *For $j \neq k$ and $0 \leq \varepsilon \leq w_j$, let $\mathbf{w}' = \mathbf{w} + \varepsilon(\mathbf{e}_k - \mathbf{e}_j)$. Then*

$$\mathrm{RIE}(\mathbf{w}') = \mathrm{RIE}(\mathbf{w}) + \varepsilon\,(g_k - g_j) - \varepsilon^2 |r_{jk}|.$$

Proof. Expand the quadratic form: $\frac{1}{2}(\mathbf{w} + \varepsilon\mathbf{d})^\top \tilde{R}(\mathbf{w} + \varepsilon\mathbf{d}) = \mathrm{RIE}(\mathbf{w}) + \varepsilon\,\mathbf{d}^\top \tilde{R}\mathbf{w} + \frac{1}{2}\varepsilon^2\,\mathbf{d}^\top \tilde{R}\mathbf{d}$ with $\mathbf{d} = \mathbf{e}_k - \mathbf{e}_j$. Here $\mathbf{d}^\top \tilde{R}\mathbf{w} = g_k - g_j$ and $\mathbf{d}^\top \tilde{R}\mathbf{d} = \tilde{R}_{kk} + \tilde{R}_{jj} - 2\tilde{R}_{jk} = -2|r_{jk}|$. □

Corollary 1. *Moving weight from criterion j to criterion k lowers RIE whenever $g_k \leq g_j$ and $r_{jk} \neq 0$: weight should flow towards criteria that are less correlated with the* currently weighted *criteria. This replaces the manuscript's "Property 3", which changed one product $w_j w_k$ in isolation; that is not possible on the simplex.*

Proposition 3 (Invariances)**.** *RIE, WMR, HHI and H are invariant to (i) relabelling the criteria and (ii) reversing the direction of any criterion (cost ↔ benefit), since reversal changes the sign of r_{jk} but not $|r_{jk}|$.*

Proof. (i) Permuting rows and columns of $\tilde{R}$ together with the entries of $\mathbf{w}$ leaves the quadratic form unchanged. (ii) Reversing criterion j multiplies row and column j of R by -1. □

Proposition 4 (Equal-weight baseline)**.** *Let $\bar{r} = \frac{1}{n(n-1)} \sum_{j \neq k} |r_{jk}|$. Then $\mathrm{RIE}(\mathbf{1}/n) = \frac{n-1}{2n}\,\bar{r}$, and $\mathrm{RIE}_{\mathrm{norm}}(\mathbf{w}) = \mathrm{RIE}(\mathbf{w})/\mathrm{RIE}(\mathbf{1}/n)$ is defined iff $\bar{r} > 0$.*

Proof. Substitute $w_j = 1/n$: $\mathrm{RIE} = \frac{1}{2n^2} \sum_{j \neq k} |r_{jk}| = \frac{n(n-1)}{2n^2}\bar{r}$. □

Remark 2. $\mathrm{RIE}_{\mathrm{norm}} < 1$ means "less redundancy than equal weighting on the same data". It removes the dependence on n and on the overall correlation level, but it still mixes concentration and redundancy (see Proposition 6). Raw RIE values must not be compared across datasets or across different numbers of criteria.

Proposition 5 ($\mathrm{RIE}_{\mathrm{adj}}$ rewards extreme concentration)**.** *Assume $s_1 = \sum_{k \neq 1} |r_{1k}| > 0$ and let $\mathbf{w}_\varepsilon = (1 - \varepsilon)\mathbf{e}_1 + \frac{\varepsilon}{n-1} \sum_{k \neq 1} \mathbf{e}_k$. Then*

$$\mathrm{RIE}_{\mathrm{adj}}(\mathbf{w}_\varepsilon) \;\sim\; \frac{s_1}{(n-1)\,\log(1/\varepsilon)} \;\longrightarrow\; 0 \qquad (\varepsilon \to 0).$$

Proof. $\mathrm{RIE}(\mathbf{w}_\varepsilon) = (1-\varepsilon)\frac{\varepsilon}{n-1} s_1 + \frac{\varepsilon^2}{(n-1)^2} \sum_{1<j<k} |r_{jk}| = \frac{s_1}{n-1}\varepsilon + O(\varepsilon^2)$, and $H(\mathbf{w}_\varepsilon) = -(1-\varepsilon)\log(1-\varepsilon) - \varepsilon \log \frac{\varepsilon}{n-1} = \varepsilon \log(1/\varepsilon) + O(\varepsilon)$. Divide. □

Remark 3. Hence $\mathrm{RIE}_{\mathrm{adj}}$ does *not* penalise sparsity: putting almost all weight on one criterion drives it below any balanced vector. On QWS, $\varepsilon = 10^{-3}$ already gives $\mathrm{RIE}_{\mathrm{adj}} = 0.0125$, lower than Entropy-TOPSIS (0.0165), the most concentrated method in the study. The convergence is slow (logarithmic), which is why moderate concentration appears to be penalised. $\mathrm{RIE}_{\mathrm{adj}}$ should be withdrawn.

Proposition 6 (Decomposition into concentration and redundancy)**.** *For every $\mathbf{w}$ with $\mathrm{HHI}(\mathbf{w}) < 1$,*

$$\mathrm{RIE}(\mathbf{w}) = \underbrace{\tfrac{1}{2}\big(1 - \mathrm{HHI}(\mathbf{w})\big)}_{\textit{spread of weights}} \times \underbrace{\mathrm{WMR}(\mathbf{w})}_{\textit{redundancy of weighted pairs}}, \qquad 0 \leq \mathrm{WMR}(\mathbf{w}) \leq 1,$$

and $\mathrm{WMR}(\mathbf{1}/n) = \bar{r}$. Moreover $\min |r_{jk}| \leq \mathrm{WMR}(\mathbf{w}) \leq \max |r_{jk}|$, where min and max run over pairs $j \neq k$ with $w_j w_k > 0$.

Proof. The identity follows from $\sum_{j \neq k} w_j w_k = 1 - \mathrm{HHI}$ and $2\,\mathrm{RIE} = \sum_{j \neq k} w_j w_k |r_{jk}|$. WMR is a weighted average of the $|r_{jk}|$ with non-negative weights $w_j w_k$ summing to $1 - \mathrm{HHI} > 0$, which gives the range. At uniform weights all pair weights are equal, so WMR is the plain mean $\bar{r}$. □

Remark 4 (Recommended reporting). WMR is bounded in $[0, 1]$ and measured in units of correlation, so unlike RIE it can be compared across datasets and numbers of criteria. We recommend reporting WMR (redundancy) together with $N_{\mathrm{eff}} = 1/\mathrm{HHI}$ (concentration) instead of $\mathrm{RIE}_{\mathrm{adj}}$.

3 Application to QWS

Table 1: RIE decomposed into its two parts on QWS (Option A BWM vectors).

Method	RIE	$\frac{1}{2}(1 - \mathrm{HHI})$	WMR	N_{eff}
Equal	0.0644	0.4444	0.1449	9.00
SD	0.0620	0.4348	0.1426	7.67
CRITCLOUD ($\alpha = 0.5$)	0.0582	0.4383	0.1327	8.11
CRITIC	0.0571	0.4325	0.1321	7.41
BWM	0.0567	0.4251	0.1334	6.68
MEREC	0.0362	0.3674	0.0985	3.77
Entropy	0.0218	0.3200	0.0682	2.78

The decomposition shows that Entropy's low RIE has two causes: it is concentrated ($N_{\mathrm{eff}} = 2.78$) *and* it weights weakly correlated criteria (WMR = 0.068). CRITIC and CRITCLOUD lower redundancy only slightly relative to equal weighting (WMR 0.132–0.133 vs 0.145) while keeping 7–8 effective criteria.

Remark 5 (Scope). None of these results links RIE or WMR to ranking accuracy. In the ground-truth simulation, lower RIE is not consistently associated with higher accuracy (results file S9). Both quantities are *descriptive diagnostics* of a weight vector, not measures of decision quality.